



Architecture and Testing of AI Tutors that Remember Past Interactions for Language Learning

Bachelor's Project Thesis

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Abstract: Traditional language learning approaches struggle with providing adaptive interactions while maintaining effective memory management for long-term retention. This thesis presents a conversational AI system with memory features that integrates spaced repetition principles with real-time conversational learning. We experimentally compared the conversational agent condition against traditional flashcard learning using vocabulary recall performance and motivational assessment. Results failed to detect significant difference in learning outcomes between conditions (conversational: 62.7%; flashcard: 70.9%; $p = .281$, $r = 0.357$). However, conversational agents generated higher learner attention and satisfaction scores with moderate effect sizes ($r \approx 0.40$), indicating users preferred it even though learning outcomes were similar. Individual differences emerged as the primary factor, with performance variations ranging from -50% to +30% between conditions for individual participants.

1 Introduction

AI chatbots increase engagement and provide valuable speaking practice (Belda-Medina & Calvo-Ferrer, 2022; Du & Daniel, 2024), offering personalized language learning experiences for many users. Despite these benefits, current systems face challenges in combining natural conversational interaction with systematic vocabulary acquisition. While incidental learning through conversation can be 1.7 times (in short-term studies) to 12 times (in long-term studies) more efficient per minute than direct instruction (McQuillan, 2019), to be this efficient, the system must handle both live conversation and help users remember words long-term. Empirical evidence demonstrates that optimal spacing can improve vocabulary recall by 24.7 percentage points, with corrective feedback providing an additional, smaller boost (Belardi et al., 2021), yet implementing these principles within natural spoken conversation remains technically challenging.

Theoretical foundations. Four theories of second language acquisition establish requirements for effective vocabulary learning systems. Skill Acquisition Theory (DeKeyser, 2007) describes the

transformation from declarative knowledge (knowing word meanings) to procedural knowledge (using words fluently), requiring both explicit introduction and contextual practice. The Interaction Hypothesis (Long, 1996) emphasizes meaning negotiation during conversation as the mechanism for connecting linguistic forms to meanings. The Noticing Hypothesis (Schmidt, 1990) establishes that conscious attention to forms is necessary for acquisition—mere exposure is insufficient. The New Theory of Disuse (Bjork & Bjork, 1992) distinguishes between storage strength (long-term consolidation) and retrieval strength (current accessibility), explaining the paradox of well-learned words becoming temporarily inaccessible without practice.

Current limitations. Existing systems fail to bridge the gap between structured learning and natural conversation. QuizBot (Li et al., 2019), a text-based quiz agent, achieved 2.6 times greater engagement than flashcards and learning improvements of over 20%, but operates through discrete question-answer exchanges rather than spontaneous dialogue. Users type responses to prompts instead of practicing real-time spoken vocabulary use. This limitation; the absence of authentic voice

conversation, prevents development of procedural fluency required for actual communication.

Current voice conversational agents face a different problem: they cannot systematically track and reintroduce vocabulary. Most agents rely solely on conversation context without persistent memory of individual word learning histories. When systems attempt to combine conversation with spaced repetition algorithms (Settles & Meeder, 2016), they typically interrupt dialogue flow for scheduled reviews, destroying conversational authenticity. No existing system successfully maintains natural spoken interaction while teaching vocabulary in an organized way.

Proposed approach. This work explores whether distributing learning requirements across specialized agents enables authentic conversational vocabulary acquisition. Unlike quiz-based systems, this approach keeps conversation natural while teaching vocabulary step-by-step. The multi-agent architecture allows each component to focus on specific theoretical requirements, one agent maintains dialogue flow while another tracks learning progress.

The system was developed in two versions:

1. A **primary version** using Neo4j (a graph database) knowledge graphs for vocabulary representation with spaced-repetition scheduling based on forgetting curves, incorporating spacing intervals validated by Belardi et al. (2021).
2. An **experimental version** for controlled testing, using simplified CSV-based storage and priority scheduling to fit experimental constraints.

Research Question. How does vocabulary acquisition through multi-agent voice conversational practice compare to traditional flashcard methods in terms of learning outcomes and learner motivation?

A within-subjects experiment compares the conversational system with traditional flashcard practice for C2-level English vocabulary acquisition, measuring both recall performance and intrinsic motivation using the RIMMS scale (Loorbach et al., 2015).

Contributions of this study include:

1. **Implementation of a theory-grounded voice conversational language learning system:** Development and testing of a multi-agent approach that integrates SLA principles through authentic spoken dialogue rather than text-based quizzing.
2. **Experimental validation:** Empirical comparison of voice conversational vocabulary practice with traditional flashcard methods, providing evidence for dialogue-based vocabulary learning effectiveness.

2 System Design

2.1 Architecture Overview

Building on the theoretical requirements outlined in the introduction, the system divides vocabulary learning between three different agents. Figure 2.1 illustrates their coordination: the Realtime Conversation Agent handles user interaction, the Language Tutor Supervisor manages vocabulary selection and scheduling, and the Language Processor tracks linguistic performance.

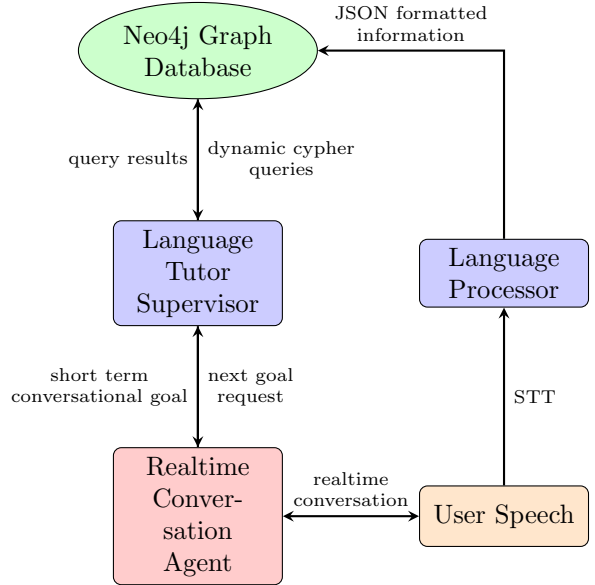


Figure 2.1: System architecture employed for guided conversation in language learning.

The architecture creates a feedback loop: the conversation agent requests learning objectives from the supervisor, which queries the graph database for appropriate vocabulary based on spaced repetition intervals and learner progress. These objectives return as conversational goals integrated naturally into dialogue. Simultaneously, the Language Processor analyzes interactions to identify successful usage, errors, and retrieval failures, updating the knowledge graph to inform future vocabulary selection.

2.2 Core Components

2.2.1 Realtime Conversation Agent

Uses gpt-4o-mini-realtime for fast, affordable conversation while incorporating learning objectives from the supervisor. Includes target words without making conversation feel unnatural, supporting procedural knowledge development through meaningful contexts.

2.2.2 Language Tutor Supervisor

Employs gpt-4o through function calls to other systems for complex reasoning tasks. Receives goal requests (e.g., "user wants to study vocabulary X"), generates database queries (using Neo4j's Cypher language), and formulates text-based learning goals for dialogue integration.

2.2.3 Language Processor

Analyzes conversation text to output organized data about each word (in JSON format) and its grammatical context. Identifies successful deployments, morphological errors, and retrieval failures without interrupting dialogue flow.

2.3 Knowledge Graph and Memory System

2.3.1 Graph Structure

The Neo4j database models vocabulary learning through interconnected nodes. **Lexeme** nodes represent vocabulary items and connect to **GrammarContext** nodes via **USES_GRAMMAR** relationships. Grammar contexts link to morphological features through **HAS_CASE**, **HAS_TENSE**, and

HAS_NUMBER relationships, enabling detection of systematic grammatical patterns.

The core learning data resides in **LearningProgress** nodes connected through:

(User)-[:HAS_PROGRESS]->

(LearningProgress)-[:ABOUT]->(Lexeme)

These nodes unify spaced repetition scheduling with performance analytics.

2.3.2 Learning Progress Integration

Each **LearningProgress** node contains:

- **Scheduling data:** SRS level (1-5 Leitner box system), next review date, last interaction timestamp
- **Performance metrics:** Overall success rates, form-specific statistics (JSON), weakest forms array, cumulative encounter/success counts

This structure uses the storage-retrieval strength distinction from the New Theory of Disuse (Bjork & Bjork, 1992). The system classifies interactions as accurate contextual usage, morphological errors, or retrieval failures, immediately updating both metrics and scheduling parameters to ensure decisions reflect actual learning outcomes.

2.3.3 Adaptive Path Generation

The integrated data enables the system to adjust what to teach based on timing and grammar skills. Exponential intervals (1, 2, 4, 8, 16 days) adjust based on performance: words with low retrieval strength receive frequent practice through SRS level reduction, while high-retrieval items maintain longer intervals for consolidation. When users master specific forms (e.g., nominative case), the system introduces related forms (e.g., accusative) using familiar lexemes, prioritizing areas identified in the **weakestForms** array.

2.4 Technical Implementation

The system builds on the April 2025 OpenAI Realtime Agents demo (Swarm), restructured to accommodate continuous linguistic analysis. Model allocation balances cost and capability: gpt-4o-mini-realtime for responsive conversation, gpt-4o for complex learning decisions. The Language Tutor Supervisor agent uses custom function calls

(prewritten functions that can be executed by LLM agents) to other systems for Neo4j integration and real-time processing. Worth noting, the June 2025 update to Swarm added a similar overseeing agent by default, but the current implementation predates this update, with a distinctly different implementation.

2.5 Experimental Adaptation

Single-session constraints necessitated architectural modifications while preserving core multi-agent coordination. The adaptation maintained task specialization but replaced graph-based persistence with CSV storage and priority-queue scheduling experimental validation.

2.5.1 Two-Agent Handoff System

The experimental implementation divided vocabulary learning between two voice agents using OpenAI Swarm’s handoff capability (Figure 2.2). The `IntroducerAgent` presented new vocabulary with definitions, etymology, and contextual examples—optimized for single-word introduction to control cognitive load. The `ReviewAgent` practiced previously introduced vocabulary through varied contexts, handling multiple words simultaneously for efficient reinforcement.

Participants controlled learning flow through verbal commands such as “I want new words” or “I want to review words”, which triggered agent handoff. This separation allowed for distinct learning direction at the request and control of the user.

2.5.2 Simplified Coordination

Direct function calls (function execution by the agent to functions, `getNewWords` and `getReviewWords`) replaced the Language Tutor Supervisor’s database query generation. The `IntroducerAgent`’s function retrieved the next unIntroduced word; the `ReviewAgent`’s function selected the least-reviewed introduced word for distributed practice. The `LanguageProcessor` continued analyzing conversations but generated CSV updates tracking usage frequency, successful deployments, and error patterns for each word in the 20-word experimental sets.

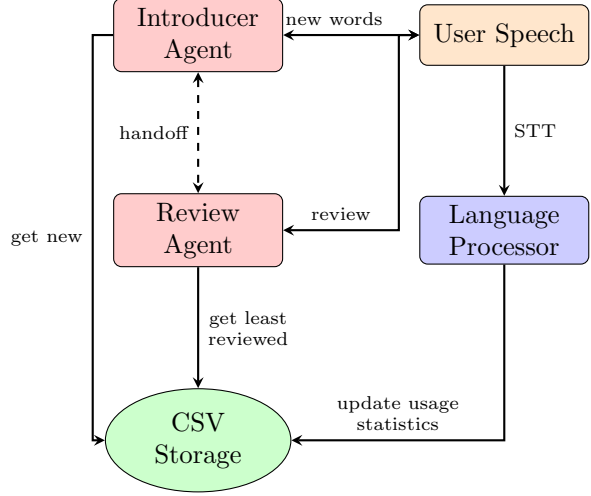


Figure 2.2: Experimental adaptation with two-agent handoff system. Verbal commands trigger transitions between introduction and review modes, with direct function calls to other systems replacing complex supervisor coordination.

2.5.3 Priority Queue Scheduling

Given that spaced repetition optimally requires weeks to demonstrate effectiveness, the experimental system implemented priority-based scheduling in which recently introduced words with lowest user frequency received priority for review, and unseen words for introduction. This maintained adaptive word selection within the 6-minute blocks while validating the multi-agent approach.

3 Methods

3.1 Overview

This study employs a within-subjects experimental design comparing memory-augmented conversational agents with traditional flashcard learning for vocabulary acquisition. We assessed both word recall performance and learner motivation using a reduced Instructional Materials Motivation Survey (RIMMS) (Loorbach et al., 2015). The experiment used the experimental adaptation of the system architecture (Section 2, subsection 2.5) adapted for single-session evaluations.

3.2 Experimental Design

We employed a counterbalanced within-subjects design with two conditions:

- **Condition A (CONV):** Interaction with the conversational agent system using the simplified implementation described above
- **Condition B (Control):** Traditional flash-card learning with definitions and 1-2 contextual examples per word

Each participant learned 20 words total (10 per condition) across four 6-minute blocks with 30-second breaks between blocks. Participants were systematically assigned to either ABAB or BABA order based on participant number (odd participants: ABAB; even participants: BABA). The 6-minute block duration was selected to enable four learning blocks plus assessments within a 35-minute total experiment time, balancing sufficient exposure with practical session constraints. RIMMS assessments were administered after the third and fourth blocks to balance memory spacing with survey burden.

3.3 Participants

Twelve university students were initially recruited, all second-language English speakers with proficiency levels primarily at C1-C2 according to the Common European Framework of Reference (CEFR). One participant (P001) was excluded from analysis due to incomplete posttest data following an initial format issue that was subsequently corrected.

The final analysis included eleven participants. The alternating assignment pattern resulted in 5 participants in ABAB order and 6 in BABA order, creating a slight imbalance due to the odd number of participants.

3.4 Materials

Vocabulary Selection Twenty-nine C2-level English words plus one control word ("happy") formed the experimental corpus. Sample vocabulary included: obfuscate, disparage, perfunctory, precocious, circumspect, capitulate, vociferous, intractable, abrogate, abstruse, acumen, admonish, austere, benevolent, bolster, cacophony, cajole,

candor, capricious, complacent, conciliatory, conundrum, copious, cursory, deleterious.

Pretest Tool A GUI-based pretest presented all 30 vocabulary items (29 C2-level words plus the control word "happy"). Participants selected words they already knew, which were removed from their learning set. An attention check required selection of "happy" to ensure task engagement. Participants knowing more than 9 of the 29 C2-level words (leaving fewer than 20 unknown words) shall be excluded from the study to ensure sufficient learning opportunity. After the pretest, Experimental Master Controller automatically selected 20 words from their unknown vocabulary and randomly assigned 10 to each experimental condition.

Experimental Master Controller A custom software tool was developed to automate and standardize the entire experimental procedure. This controller eliminated manual administration tasks and ensured identical experimental conditions for all participants. The software handled all aspects of the experiment: administering the vocabulary pretest, automatically generating personalized 20-word learning sets based on pretest results, loading the correct word lists into each learning system at the appropriate blocks, enforcing precise timing for all 6-minute learning blocks and 30-second breaks, triggering RIMMS surveys at predetermined points, and generating customized posttest questions unique to each participant's assigned vocabulary. By automating these tasks, the controller reduced experimenter workload, eliminated timing errors, and ensured that each participant experienced exactly the same experimental flow regardless of when they participated or which researcher conducted the session.

3.5 Measures

Vocabulary Assessment A multiple-choice posttest administered 12-24 hours post-session via Google Forms tested all 20 learned words. Questions assessed contextual understanding, for example: "The politician tried to ____ the facts to avoid taking responsibility for the scandal" with "obfuscate" among the options. The assessment was automatically generated based on each participant's personalized vocabulary set.

Motivation Assessment The reduced RIMMS measured four dimensions (Attention, Relevance, Confidence, and Satisfaction) using 5-point Likert scales. Each dimension was assessed with targeted questions administered immediately after participants completed their second exposure to each condition.

3.6 Procedure

The experiment was conducted in-person in a quiet environment. Upon arrival, participants were seated at a computer workstation with the experimental software preloaded. The experimenter remained present throughout the session to ensure protocol adherence and address technical issues, but did not interact with participants during learning blocks.

After obtaining written informed consent, the experimenter provided standardized instructions explaining the experimental structure and both learning systems. For the conversational agent condition, participants were specifically instructed to control their learning by verbally requesting either "I want new words" or "I want to review words" to switch between vocabulary introduction and practice modes. Participants were encouraged to actively direct their learning pace and strategy using these commands.

The experimental session followed the structure shown in Figure 3.1. Participants first completed the vocabulary familiarity pretest, identifying known words from the C2 corpus. The Experimental Master Controller then generated their personalized 20-word learning set and assigned words to conditions.

Learning blocks proceeded with automated transitions and timing enforced by the Master Controller. Visual and auditory signals indicated block transitions. During CONV blocks, participants engaged with the conversational agent through the web interface, while Control blocks presented flashcards with navigation controls.

The experimenter then debriefed participants, reminding to take the posttest the next day, answered any questions.

The vocabulary posttest was automatically sent via email 12 hours after session completion. Participants completed this assessment at home without

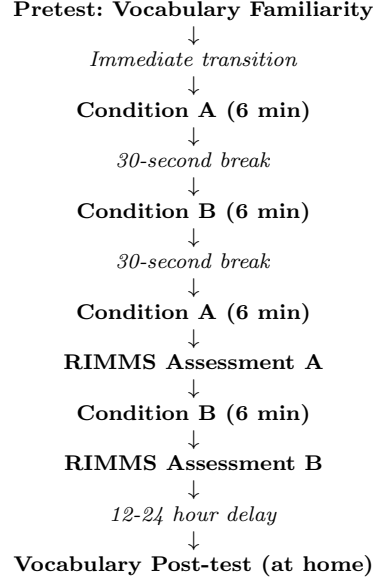


Figure 3.1: Complete experimental flow for ABAB condition order. BABA participants experienced the same structure with conditions reversed. Total session duration: approximately 35 minutes.

supervision, using their personal devices. They had a 12-hour window to complete the assessment.

3.7 Data Analysis

Performance Analysis Learning effectiveness was assessed through delayed recall accuracy on the multiple-choice posttest. Due to small sample size ($n=11$), discrete performance data (10% increments resulting from 10 questions per condition), and non-normal distributions confirmed by Shapiro-Wilk tests, we employed Wilcoxon signed-rank tests to compare conditions. Effect sizes were calculated using $r = z/\sqrt{n}$ to quantify the magnitude of differences.

Motivation Analysis RIMMS responses were analyzed using Wilcoxon signed-rank tests for dimension-level comparisons between conditions. Additionally, correlation analysis examined relationships between motivational dimensions and performance outcomes to identify potential mediating factors.

Statistical Power Post-hoc power analysis revealed 34% power for the observed performance difference ($r = 0.31$) and 54% power for motivational effects ($r \approx 0.40$). Approximately 21 participants would be required to achieve 80% power for detecting moderate effects. This underpowered status suggests cautious interpretation of non-significant findings, as meaningful differences may exist but were not detectable with the current sample size.

4 Results

This section presents experimental findings from eleven participants who completed both vocabulary learning conditions, analyzing performance outcomes and motivational responses.

4.1 Participant Characteristics

Eleven participants successfully completed the full experimental protocol, including vocabulary pretest, both learning sessions in counterbalanced order, RIMMS motivation assessments, and the 12-24-hour delayed posttest. Five participants received the ABAB condition order and six received the BABA order. All participants were C1-C2 level English speakers who knew fewer than 9 of the 29 target vocabulary words during pretesting, ensuring sufficient learning opportunity.

4.2 Learning Performance Outcomes

Overall Performance Comparison The 12-24 hour delayed vocabulary assessment revealed comparable retention rates between learning conditions. Conversational learning achieved a mean accuracy of 62.7% (SD = 21.4%, Range: 20%-90%), while flashcard learning yielded 70.9% (SD = 17.3%, Range: 30%-100%). Although the 8.2 percentage point difference favored flashcards, this difference did not reach statistical significance.

Given the small sample size ($n=11$), discrete performance measurements (10% increments), and presence of multiple tied values, we employed non-parametric analysis. A Wilcoxon signed-rank test confirmed no significant difference between conditions ($W = 7.0$, $p = .281$, $r = 0.357$), indicating that both methods produced equivalent learning

outcomes under our experimental conditions (Figure 4.1).

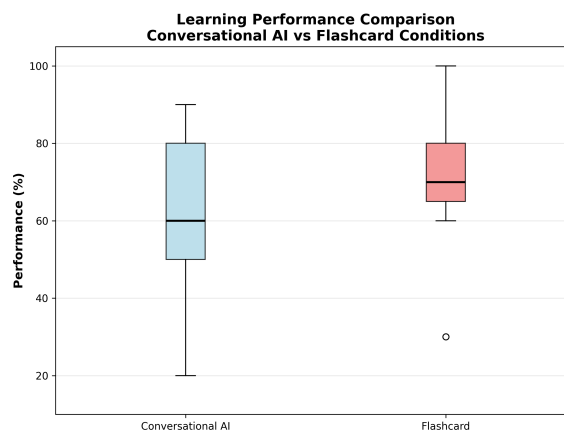


Figure 4.1: Learning performance comparison between conversational AI and flashcard conditions. Box plots show medians, quartiles, and distributions with means marked as red diamonds. Statistical analysis revealed no significant difference between conditions (Wilcoxon $W = 7.0$, $p = .281$, $r = 0.357$).

Individual Learning Profiles While aggregate performance was similar, individual responses revealed striking differences in method effectiveness. Performance differences between conditions for individual participants ranged from -50% to +30%, indicating distinct learning profiles rather than universal method superiority.

The data revealed three distinct learner profiles: conversational learners ($n=3$; P003, P006, P007) achieved 20-30% higher scores with the conversational agent; traditional learners ($n=4$; P004, P008, P009, P010) performed 30-50% better with flashcards, with P010 achieving perfect recall; and method-agnostic learners ($n=4$; P002, P005, P011, P012) demonstrated identical 70-80% performance regardless of method. This three groups of learners show that the best method depends on the individual (Figure 4.2).

4.3 Motivational Response Analysis (RIMMS)

The Reduced Instructional Materials Motivation Survey revealed different effects on the four types

Individual Performance Differences Between Learning Conditions

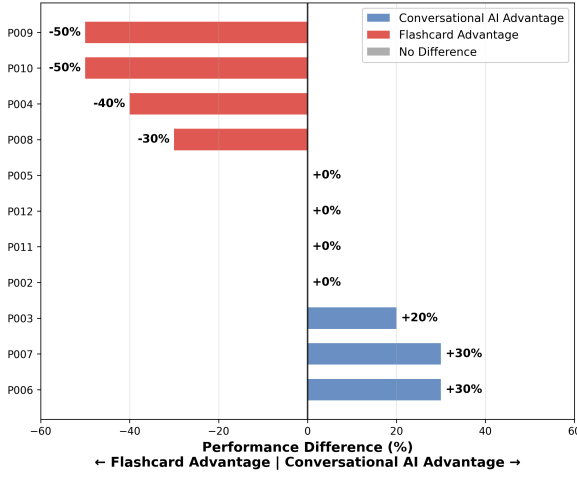


Figure 4.2: Individual performance differences between learning conditions. Horizontal bars extend from center showing the magnitude and direction of performance differences. Red bars indicate flashcard advantages, blue bars show conversational AI advantages. Participants are sorted by difference magnitude, revealing distinct individual learning preferences ranging from -50% to +30%.

of motivation. While statistical significance was not achieved due to sample size constraints, moderate effect sizes suggest meaningful practical differences.

Attention and Engagement Conversational agents demonstrated superior attention capture, generating mean attention scores of 3.88 (SD = 1.15) compared to 2.85 (SD = 1.13) for flashcards. Although the Wilcoxon signed-rank test did not reach significance ($W = 17.5$, $p = .183$), the moderate effect size ($r = 0.402$) suggests that conversational interaction may be more effective at maintaining learner focus during vocabulary acquisition.

Satisfaction and User Experience A similar pattern emerged for satisfaction, with conversational learning producing higher ratings ($M = 3.70$, $SD = 1.00$) than flashcard learning ($M = 2.91$, $SD = 1.15$). While not statistically significant ($W = 14.0$, $p = .188$), the moderate effect size ($r = 0.405$) indicates a meaningful trend toward enhanced user experience with conversational agents.

Relevance and Confidence Both methods were perceived as equally relevant to learning goals (Conversational: $M = 4.03$, $SD = 0.52$; Flashcard: $M = 3.94$, $SD = 0.55$; $W = 17.0$, $p = .961$, $r = 0.015$). Similarly, learner confidence showed minimal differences between conditions (Conversational: $M = 3.58$, $SD = 0.88$; Flashcard: $M = 3.33$, $SD = 0.57$; $W = 13.0$, $p = .539$, $r = 0.168$). These findings suggest that while conversational agents enhance engagement, they do not necessarily increase perceived relevance or learning confidence (Figure 4.3).

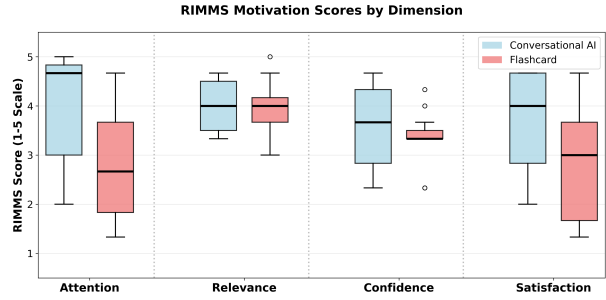


Figure 4.3: RIMMS motivation score distributions across four dimensions comparing conversational AI and flashcard conditions. Box plots show medians, quartiles, and outliers for each condition. Conversational AI demonstrated higher attention and satisfaction scores, while relevance and confidence showed minimal differences.

4.4 Motivation-Performance Relationship

Exploratory correlation analysis revealed distinct patterns between motivational dimensions and learning outcomes across conditions. For conversational learning, motivation showed weak associations with performance (attention-performance: $r = .35$, $p = .293$; satisfaction-performance: $r = .33$, $p = .315$). Flashcard learning demonstrated somewhat stronger relationships, particularly between attention and performance ($r = .54$, $p = .089$), which approached statistical significance.

These differential correlation patterns suggest that motivation may play different roles in each learning method. The stronger attention-performance relationship for flashcards implies that

sustained focus is more critical for success with traditional methods, while conversational learning may engage learners through mechanisms beyond conscious attention. However, these exploratory findings require replication with larger samples before drawing firm conclusions.

5 Discussion

5.1 Methodological Constraints and Their Implications

This pilot study’s findings must be interpreted within significant methodological constraints. With only 11 participants and 34% statistical power, we cannot claim equivalence between methods, only that we failed to detect differences. Several confounding factors likely influenced the observed performance patterns.

The most significant limitation was the training asymmetry between conditions. Participants brought years of flashcard experience but received no prior training for AI conversational learning. Despite pre-experiment instructions, participants required time during the experimental sessions to learn the conversational system’s operation. This interface familiarization occurred within the fixed 6-minute blocks allocated for vocabulary learning, reducing time available for vocabulary exposure. The flashcard condition required no such adaptation period, allowing immediate focus on vocabulary. The conversational condition’s comparable outcomes (62.7% vs. 70.9%, $p = .281$) despite these disadvantages may indicate untapped potential.

Our assessment captured only declarative knowledge through multiple-choice recognition tests, potentially missing advantages in productive vocabulary use. The 24-hour delay, while standard for flashcard studies, may not capture differential forgetting rates between rote memorization and contextual encoding. Without tracking exposure frequency per word or measuring when participants achieved interface proficiency, we cannot determine whether performance differences reflect method effectiveness or simply exposure quantity.

5.2 Tentative Findings Within Constraints

Given these limitations, our results provide preliminary evidence for patterns requiring future validation. Performance differences between conditions for individual participants ranged from -50% to $+30\%$. With each participant representing 9% of our sample, these variations could reflect noise rather than stable preferences. We observed three tentative patterns: three participants (27%) showed advantages with conversational agents, four (36%) performed better with flashcards, and four (36%) showed similar performance across conditions. These observations suggest hypotheses about learner-method fit rather than established profiles.

Conversational agents generated higher attention ($r = 0.40$) and satisfaction ($r = 0.41$) scores, though these moderate effects did not reach statistical significance in our small sample. The possibility that sustained engagement influences long-term learning persistence remains unexplored in our brief study.

5.3 Theoretical Evaluation Under Experimental Constraints

Our experimental design could not adequately test the theoretical frameworks motivating this work. The assessment evaluated only declarative knowledge (word recognition), not the procedural knowledge (word use) that Skill Acquisition Theory suggests conversational practice develops. Without production tasks or exposure metrics, we cannot evaluate whether method differences reflect learning quality or quantity.

The Interaction Hypothesis requires sustained meaning negotiation through conversational interaction. Our 12-minute total conversational exposure per participant was insufficient to test Long’s (1996) predictions, which assume weeks of varied contextual encounters rather than minutes of initial exposure. Similarly, the storage-retrieval strength predictions from the New Theory of Disuse operate on timescales exceeding our 24-hour assessment window. Testing whether conversational agents maintain retrieval strength while flashcards build storage strength requires longitudinal designs tracking forgetting curves over multiple weeks.

The ARCS Model received partial validation

through selective enhancement of attention and satisfaction without confidence gains. While conversational agents successfully captured attention and generated satisfaction, unchanged confidence indicates that engagement alone doesn't translate to perceived mastery. This dissociation suggests that system design must explicitly support competency development beyond engagement.

5.4 Implications and Future Directions

Rather than making design recommendations from underpowered findings, we identify methodological requirements for future research. Valid comparisons require minimum 2-week durations to assess memory consolidation patterns, structured training protocols equalizing method familiarity, production-based assessments capturing vocabulary use beyond recognition, and sample sizes providing 80% power (approximately $n = 21$ based on observed effect sizes). Studies should also quantify exposure differences between conditions to distinguish method effectiveness from exposure quantity.

Our preliminary observations suggest hypotheses worth pursuing. The suspected exposure differences between conditions merit systematic investigation through tracking word presentation frequency. If conversational learning achieves comparable outcomes with fewer exposures, this would support theoretical predictions about contextual encoding's value. The observed individual variations, while not statistically reliable, suggest that learner-method fit may be more important than universal method superiority. The engagement advantages of conversational agents might influence voluntary practice patterns and long-term retention in ways our brief study could not detect.

5.5 Contributions Despite Limitations

This pilot study makes several contributions despite its constraints. We demonstrated that multi-agent coordination can preserve conversational fluency while implementing vocabulary tracking and reintroduction. The system successfully maintained natural dialogue while pursuing learning objectives, validating the technical approach even if learning benefits across all individuals remain unproven.

Our experience reveals requirements for evaluating conversational learning systems. The training asymmetry problem, interface learning time, and assessment limitations we encountered provide guidance for future research design. While our sample size prevents strong claims, the individual learning-style variations observed (if replicated) would challenge one-size-fits-all approaches to educational technology deployment.

5.6 Conclusions

This pilot investigation of memory-augmented conversational agents for vocabulary learning reveals the complexity of evaluating novel educational technologies against established methods. Our results neither support nor refute conversational agents' potential; instead, they highlight what's needed to compare methods fairly.

The comparable performance achieved by conversational agents despite methodological disadvantages suggests resilience worth investigating. The higher engagement scores, though not statistically significant with our sample size, align with theoretical predictions about conversational interaction's motivational benefits. The substantial individual variations observed, even in our small sample, suggest that asking "which method is better?" may be misguided. The more productive question appears to be "which method better serves which learners under what conditions?"

Future research should address the methodological limitations we encountered by implementing adequate training periods, controlled exposure rates, production-based assessment, and sufficient statistical power. Only with these elements can the field evaluate whether conversational AI offers advantages for vocabulary learning or simply provides an engaging alternative to proven methods.

The promise of AI in education likely lies not in completely replacing traditional methods but in giving learners more choices. Our preliminary findings suggest that some learners may benefit from conversational approaches while others are better served by traditional methods, representing an important direction for future educational technology research.

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A Code Repositories

The complete source code for this research is available in the following GitHub repositories to enable reproducibility and further development:

Primary System Implementation:

[https://github.com/coding-crying/
realtime-agents-language-tutor/tree/
main](https://github.com/coding-crying/realtime-agents-language-tutor/tree/main)

Full separation-of-concerns architecture with Neo4j graph database, spaced repetition algorithms, and OpenAI Swarm framework integration.

Experimental Version & Data Analysis:

[https://github.com/coding-crying/
S5200954-Bachelor-Project-/tree/main](https://github.com/coding-crying/S5200954-Bachelor-Project-/tree/main)

Simplified CSV-based implementation used in the experimental evaluation, including all data analysis scripts, statistical computations, visualization code, and password-protected participant data.

Each repository includes comprehensive documentation, installation instructions, usage examples, and detailed README files explaining the system architecture and experimental setup.

Attribution and Additional Materials: The base repository builds upon OpenAI’s open source framework that provides WebSocket tunnels for connecting to their Realtime Voice Agents. However, all system design, implementation, and research contributions presented in this paper were developed independently by the author. Complete GitHub merge history, additional experimental materials, and supplementary documentation are available upon request.